Seminar Report on

Corrective Retrieval Augmented Generation

(CRAG)

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Certificate

This is to certify that the seminar entitled, **"Corrective Retrieval Augmented Generation (CRAG)"**, is a bonafide work done by **"Priti Ram"** in partial fulfillment of requirements for the award of Bachelor of Technology Degree in Computer Science & Engineering (AI & ML) at **"Odisha University of Technology & Research, Bhubaneswar"** is an authentic work carried out by her under my supervision and guidance. The matter embodied in this work has not been submitted to any other University / Institute for the award of any Degree or Diploma to the best of my knowledge.

(Supervisor) H.O.S (SCS)

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Abstract

Corrective Retrieval Augmented Generation (CRAG) is an enhanced Retrieval-Augmented Generation framework designed to improve the reliability and factual accuracy of Large Language Models (LLMs). Conventional RAG systems generate responses based on retrieved documents but may produce incorrect or misleading answers when the retrieved information is irrelevant, incomplete, or unreliable. CRAG addresses this limitation by introducing a lightweight retrieval evaluator that assesses the quality of retrieved knowledge and dynamically determines an appropriate corrective action. Depending on the retrieval quality, the system can refine the retrieved information, supplement it with knowledge obtained through web search, or replace unreliable retrieval results. It further applies a knowledge refinement process to extract relevant information and reduce irrelevant content before passing the evidence to the language model. Thus, CRAG provides a more robust and adaptive approach to retrieval-based question answering, reducing the impact of retrieval errors and improving the overall quality and factuality of generated responses.

Keywords: Retrieval-Augmented Generation (RAG), Corrective Retrieval Augmented Generation (CRAG), Large Language Models (LLMs), Information Retrieval, Retrieval Evaluation, Knowledge Refinement, Hallucination Reduction, Generative AI

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1. INTRODUCTION

1.1 Background and Motivation

Large Language Models (LLMs) have demonstrated an impressive ability to understand instructions and generate fluent, coherent text. Despite this, LLMs inevitably manifest hallucinations – confidently generated statements that are factually incorrect – because the accuracy of a generated response cannot be secured solely by the parametric knowledge encapsulated in the model's weights. This limitation becomes especially problematic in knowledge-intensive tasks such as open-domain question answering, fact verification, and long-form biography generation, where up-to-date and precise factual grounding is essential.

Retrieval-Augmented Generation (RAG) has emerged as a practicable complement to LLMs. In RAG, the input to the model is augmented by prepending documents retrieved from an external knowledge corpus, allowing the generator to draw on non-parametric knowledge in addition to what it has memorized during training. While effective, RAG's performance is contingent on the relevance and accuracy of the retrieved documents. Because the retriever and the generator are tightly coupled, an unsuccessful or noisy retrieval can result in an unsatisfactory response regardless of how capable the underlying generator is. Most conventional RAG pipelines indiscriminately incorporate whatever the retriever returns, without ever asking whether those documents are actually relevant to the query.

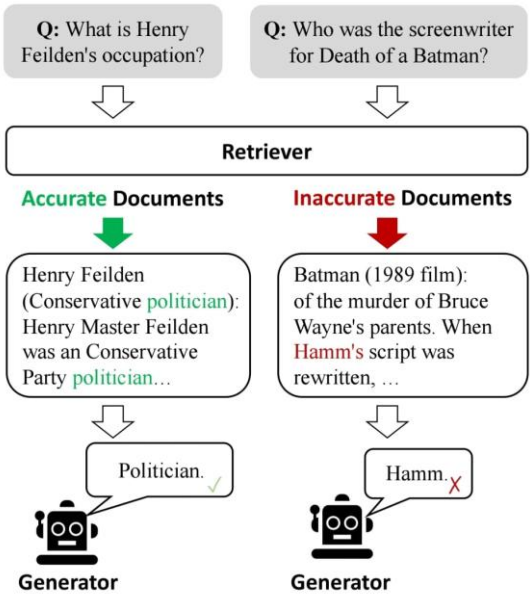


Figure 1.1: Examples showing that a low-quality retriever introduces irrelevant information that misleads the generator.

1.2 Problem Statement

This seminar is based on the paper “Corrective Retrieval Augmented Generation” by Shi-Qi Yan, Jia-Chen Gu, Yun Zhu and Zhen-Hua Ling, which specifically studies the scenario where the retriever returns inaccurate or irrelevant results. The central research question addressed is: how can a RAG system automatically recognize that its retrieved documents are of low quality, and correct itself rather than propagating irrelevant information into the generated response? Furthermore, since even a genuinely relevant document usually contains a mixture of useful and non-essential text, the paper also studies how retrieved content can be refined so that only the most critical knowledge strips are passed on to the generator.

2. LITERATURE REVIEW

The work on CRAG is grounded in several interrelated strands of prior research: studies on hallucination in LLMs, the development of Retrieval-Augmented Generation and its underlying retriever architectures, methods for evaluating and filtering the quality of retrieved evidence, more recent advanced RAG variants that try to decide when and what to retrieve, and techniques for compressing or refining retrieved context before it reaches the generator. Each of these strands is discussed in turn below, followed by a positioning of CRAG relative to this body of work.

2.1 Hallucinations of LLMs

Although LLMs exhibit impressive abilities to understand instructions and generate fluent text, one of the most persistent issues they struggle with is hallucination – the generation of content that is fluent and confident but factually incorrect or unverifiable [11], [40]. Prior surveys have identified that either outdated information or incorrectly activated parametric knowledge can seriously contribute to hallucinated output. Large-scale, loosely curated training data, an imbalance of high-quality samples, and imperfect coverage of the input space all exacerbate the problem, ultimately harming the reliability of LLM-based applications.

Researchers commonly distinguish between two broad categories of hallucination: intrinsic hallucination, where the generated content directly contradicts the source or established facts, and extrinsic hallucination, where the content cannot be verified from the available source material at all. Mitigation strategies proposed in the literature include reinforcement learning from human feedback to penalize unsupported claims [24], self-consistency and chain-of-thought prompting to encourage more deliberate reasoning, and post-hoc fact-checking pipelines that verify a generated claim against an external source after the fact [38]. A comprehensive survey catalogues a wide range of such mitigation techniques [33]. Retrieval-Augmented Generation belongs to a complementary family of solutions that intervenes before generation, by supplying relevant external evidence as part of the model's input rather than relying solely on what the model has memorized [31].

2.2 Retrieval-Augmented Generation

RAG is regarded as a useful method for addressing hallucination by enhancing the input of a generative model with documents retrieved from an external corpus such as Wikipedia [15]. This markedly improves performance on knowledge-intensive tasks such as open-domain

question answering and fact verification. The original RAG formulation combined a pre-trained dense retriever with a sequence-to-sequence generator, framing the retrieved documents as a latent variable that is marginalized over during generation [15]. A related architecture, REALM, incorporates retrieval directly into the language model's pre-training objective so that the retriever and the masked-language-modelling objective are learned jointly [9].

Underlying all of these systems is a retriever component, which is typically either sparse or dense. Sparse retrievers, such as those based on TF–IDF or BM25, rank documents by lexical overlap with the query and are computationally cheap but can miss semantically relevant documents that use different wording. Dense retrievers, by contrast, encode both queries and documents into a shared embedding space using a neural encoder – as in Contriever [10] – and retrieve based on vector similarity, which captures semantic relevance more effectively but requires a pre-built dense index. Earlier RAG studies generally treat the retriever as a fixed, trustworthy black box: the top-K documents it returns are passed to the generator largely unfiltered. This overlooks a critical question that becomes central to CRAG's motivation: what happens if the retrieval goes wrong? Prior work has shown that a model's own parametric memory is sometimes more reliable than noisy non-parametric evidence, and that irrelevant retrieved context can actively degrade generation quality [19]. If the retrieved documents are irrelevant or outdated, the retrieval system can actually exacerbate the factual errors that the language model makes, rather than correcting them, because the generator has no built-in way of knowing that the evidence it has been given is untrustworthy [37].

2.3 Evaluating and Filtering Retrieved Evidence

A smaller but growing body of work has begun to treat retrieval quality itself as something that must be actively assessed rather than assumed. It has been shown that LLMs can be easily distracted by irrelevant retrieved context, motivating the use of lightweight filters that discard passages before they reach the generator [30]. Other studies have proposed making retrieval-augmented models robust to irrelevant context directly, for instance by training the model to recognize and down-weight untrustworthy evidence rather than treating every retrieved passage as equally reliable [37]. CRAG's retrieval evaluator belongs to this line of work, but goes further by not only scoring documents but also using that score to select among three distinct downstream corrective actions, rather than simply filtering or re-ranking.

2.4 Advanced RAG Approaches: Deciding When and What to Retrieve

A number of advanced approaches have been developed to make RAG more selective and reliable by deciding adaptively whether retrieval is needed at all, rather than retrieving unconditionally for every query. Self-RAG introduces a critic model, trained jointly with the generator, that emits special reflection tokens deciding whether retrieval is needed for a given segment of text and which retrieved documents should be used, since retrieval is sometimes unnecessary or even harmful for certain queries [2]. In a similar spirit, active retrieval augmented generation anticipates upcoming content during long-form generation and triggers a new retrieval step whenever the model's confidence in its next tokens drops, so that retrieval happens exactly when it is needed rather than only once at the start [12].

Other approaches focus on how retrieved evidence is incorporated rather than when. SAIL (Search-Augmented Instruction Learning) tunes a language model on instructions with retrieved documents inserted before the instructions, teaching the model to be robust to noisy or irrelevant search results that may appear alongside genuinely useful ones [18]. Toolformer takes a different approach entirely by pre-training a model to autonomously decide when to call external tools – including a search API, a calculator, and a calendar – and how to incorporate the tool's output into its own generation, framing retrieval as one instance of a broader tool-use capability [29]. RA-ISF extends this idea further by combining retrieval with an iterative self-feedback loop, allowing a model to answer, assess its own answer, and re-retrieve as needed [17]. Other systems extend retrieval specifically to open-ended web search and dialogue, training models to issue search queries against large web corpora and incorporate the retrieved content into conversational responses [14], [25].

2.5 Knowledge Refinement and Context Compression

A separate but related line of research addresses the fact that even a genuinely relevant retrieved document is rarely uniformly useful – most of its content may be redundant, tangential, or simply too long to fit efficiently into the generator's context window. SuRe trains a mechanism that summarizes retrieved evidence with the help of candidate answers, condensing retrieved documents into a more compact and answer-relevant form before generation [13]. Other work adapts the base language model itself to make better use of a fixed retrieval pipeline, for instance by fine-tuning it to distinguish relevant from irrelevant documents during training [39], or by conditioning generation jointly on task-specific and open-world retrieved knowledge [32]. CRAG's decompose-then-recompose knowledge refinement strategy shares

this broad motivation – that raw retrieved text should be distilled before use – but implements it with the same lightweight evaluator used for the initial correctness judgment, rather than training a separate compression or summarization model, which keeps the overall pipeline simpler and cheaper to run.

2.6 Positioning of CRAG

The key distinction between CRAG and this prior work is one of focus. Approaches such as Self-RAG [2], the active retrieval strategy of Jiang et al. [12], SAIL [18], Toolformer [29] and RA-ISF [17] primarily explore whether retrieval is necessary at all, or how to use it as a general-purpose tool alongside other capabilities; relevance-filtering and robustness-focused methods [30], [37] assume the retriever has already done a reasonable job and focus on cleaning up its output; and knowledge-compression methods such as SuRe [13] focus on shortening or condensing context rather than judging its correctness. CRAG instead specifically targets the scenario in which the retriever has already returned inaccurate or entirely irrelevant results, and designs explicit, graded corrective strategies – refining internal knowledge when it is trustworthy, replacing it with web search when it is not, and blending both when the evaluator is uncertain – to recover from that failure. To the best of the authors' knowledge, this is the first work to design corrective strategies for RAG with this specific aim of improving robustness against poor retrieval, rather than treating retrieval quality as a binary switch (retrieve or don't retrieve) or as a fixed, always-trustworthy input to be merely filtered or compressed.

2.7 Proposed Solution

The authors propose Corrective Retrieval-Augmented Generation (CRAG), a method designed to self-correct the results of a retriever and improve the utilization of retrieved documents for augmenting generation. At the heart of CRAG is a lightweight retrieval evaluator that assesses the overall quality of retrieved documents for a given query and returns a confidence degree. Based on this confidence, one of three knowledge retrieval actions – Correct, Incorrect, or Ambiguous – is triggered. When the retrieved documents are judged Incorrect, large-scale web search is used as a strategic extension, since a static and limited corpus can only return sub-optimal documents in terms of scope and diversity. In addition, a decompose-then-recompose algorithm filters out redundant or irrelevant content from retrieved documents so that only key information reaches the generator.

CRAG is designed to be plug-and-play, meaning it can be seamlessly coupled with a variety of existing RAG-based approaches without requiring any modification to the underlying

generator. The authors demonstrate this adaptability by integrating CRAG with both standard RAG and the state-of-the-art Self-RAG framework [2], and by evaluating it on four datasets that cover both short-form and long-form generation tasks.

2.8 Contributions

- This work studies the scenario where the retriever returns inaccurate results and, to the best of the authors' knowledge, makes the first attempt to design corrective strategies for RAG to improve its robustness.
- A plug-and-play method named CRAG is proposed to improve the ability of automatic self-correction and efficient utilization of retrieved documents.
- Extensive experiments demonstrate CRAG's adaptability to RAG-based approaches and its generalizability across short- and long-form generation tasks.

3. METHODOLOGY

3.1 Task Formulation

Following standard RAG formulations, given an input query X and an accessible corpus of knowledge documents C = {d1, ..., dN}, the system is expected to generate an output Y. The overall framework is divided into a retriever R and a generator G. The retriever aims to retrieve the top-K documents D = {dr1, ..., drk} relevant to X from the corpus C, and the generator produces Y based on X and D. This can be formulated as $P(Y|X) = P(D|X) \cdot P(Y, D|X)$, which shows that the retriever and generator are tightly coupled; any unsuccessful retrieval can produce an unsatisfactory response irrespective of how capable the generator is. Improving the robustness of this coupling is exactly the focus of CRAG.

3.2 Overview of Model Inference

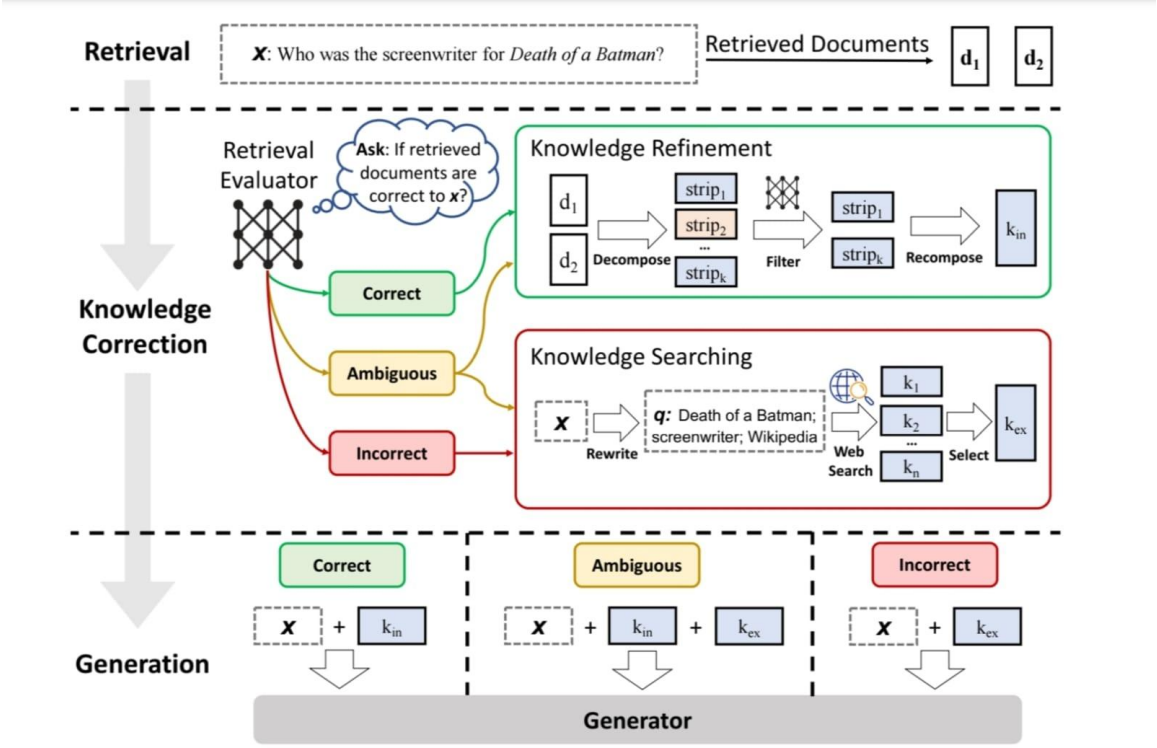


Figure 3.1: An overview of the proposed CRAG at inference

Given an input query and the documents retrieved by any off-the-shelf retriever, a lightweight retrieval evaluator first estimates the relevance score of each retrieved document with respect to the query. These scores are quantified into one of three confidence degrees, which in turn trigger a corresponding action:

- Correct: the retrieved documents are refined into more precise knowledge strips through knowledge decomposition, filtering and recomposition.
- Incorrect: the retrieved documents are discarded altogether, and a web search is performed to obtain complementary external knowledge.
- Ambiguous: a soft, balanced action that combines both refined internal knowledge and external web knowledge when the evaluator cannot confidently judge correctness.

After the retrieval results have been optimized in this way, an arbitrary generative model can be used to produce the final response, which makes CRAG plug-and-play with respect to the choice of generator.

3.3 Retrieval Evaluator

The retrieval evaluator is the component responsible for judging whether retrieved documents are accurate before they are used. It is initialized from T5-large [27] and fine-tuned for this purpose, giving it a parameter size (0.77B) that is much smaller than most current LLMs – hence the description "lightweight." For every query, typically ten documents are retrieved (using the same Contriever-based [10] results as Self-RAG [2] for fair comparison); the query is concatenated with each document individually, and the evaluator predicts a relevance score from –1 to 1 for each question–document pair. Relevance signals for fine-tuning are collected from existing datasets – for instance, PopQA [19] provides a gold subject Wikipedia title per question, which is used as a high-quality positive signal, while negative samples are drawn from retrieval results that are superficially similar to the query but not actually relevant. Notably, the authors also tried prompting ChatGPT to judge retrieval relevance directly, but found that this underperformed the dedicated fine-tuned evaluator (discussed further in Section 4.3).

3.4 Action Trigger

Two thresholds – an upper and a lower bound – are set empirically for the confidence score. A retrieval is judged Correct if the confidence score of at least one retrieved document exceeds the upper threshold, meaning that at least one relevant, reliable document exists among the results. A retrieval is judged Incorrect if the confidence scores of all retrieved documents fall below the lower threshold, meaning the entire retrieved set is considered irrelevant and unhelpful. All remaining cases, where the evaluator's judgment is not confident enough to fall cleanly into either extreme, are assigned the intermediate Ambiguous action. Preliminary

experiments using only Correct and Incorrect actions showed that CRAG's performance was overly sensitive to the accuracy of the evaluator; introducing the softer Ambiguous action significantly reduced this dependence and improved overall robustness. The complete inference procedure is summarized in Figure 3.2.

Algorithm 1: CRAG Inference

Require: E (Retrieval Evaluator), W (Query Rewriter), G (Generator)

Input: x (Input question), D = {d1, d2, ..., dk} (Retrieved documents)

Output: y (Generated response)

1 score_i = E evaluates the relevance of each pair (x, d_i), d_i ∈ D

2 **Confidence** = Calculate and give a final judgment based on {score_1, score_2, ... score_k}

// **Confidence** has 3 optional values: [CORRECT], [INCORRECT] or [AMBIGUOUS]

3 **if** Confidence == [CORRECT] **then**

4 Internal_Knowledge = Knowledge_Refine(x, D)

5 k = Internal_Knowledge

6 **else if** Confidence == [INCORRECT] **then**

7 External_Knowledge = Web_Search(W Rewrites x for searching)

8 k = External_Knowledge

9 **else if** Confidence == [AMBIGUOUS] **then**

10 Internal_Knowledge = Knowledge_Refine(x, D)

11 External_Knowledge = Web_Search(W Rewrites x for searching)

12 k = Internal_Knowledge + External_Knowledge

13 **end**

14 G predicts y given x and k

3.5 Knowledge Refinement

For documents judged relevant, a decompose-then-recompose method extracts the most critical knowledge strips. Retrieved results are segmented into internal strips: a short result of one or two sentences is treated as an individual strip, while longer documents are split into smaller

units of a few sentences each, based on total length, so that each strip is assumed to carry an independent piece of information. The same fine-tuned retrieval evaluator is then used to score each knowledge strip; irrelevant strips are filtered out (using a filter threshold), while the remaining relevant strips are recomposed via concatenation, in order, to form the internal knowledge that is passed to the generator.

3.6 Web Search for External Knowledge

When retrieval is judged Incorrect (or partially so, under Ambiguous), CRAG turns to large-scale web search as a strategic extension, since a static and limited corpus can only return sub-optimal documents in terms of scope and diversity [25]. The original query is first rewritten by ChatGPT, using a few-shot prompt, into at most 3 keywords that mimic how a person would phrase a web search (for example, "What is Henry Feilden's occupation?" becomes "Henry Feilden, occupation"). A commercial search API is then used to obtain a ranked list of URLs, with authoritative sources such as Wikipedia preferred to reduce the risk of bias or unreliable content. The content of these web pages is transcribed and passed through the same decompose-then-recompose knowledge refinement pipeline described above to produce the external knowledge used for generation.

3.7 Datasets, Metrics and Baselines

CRAG is evaluated on four public datasets covering different task types: PopQA [19] (short-form entity generation, measured by accuracy on a long-tail subset of 1,399 rare-entity queries), Biography (long-form generation, measured using FactScore [20]), PubHealth (true-or-false health claim verification, measured by accuracy) and Arc-Challenge [4] (multiple-choice science questions, measured by accuracy). CRAG is compared against baselines without retrieval (e.g., LLaMA2 [35], Alpaca, CoVE [7]), baselines with standard retrieval (e.g., LLaMA2 and Alpaca with top retrieved documents prepended), and advanced RAG approaches such as SAIL [18] and Self-RAG [2] and proprietary LMs trained with private retrieval-augmented data, namely Ret-ChatGPT, Ret-LLaMA2-chat, and perplexity.ai, an InstructGPT-based production search system. To demonstrate CRAG's plug-and-play adaptability, it is integrated into standard RAG (yielding a model referred to as CRAG) as well as into Self-RAG (yielding Self-CRAG).

4. RESULTS

4.1 Overall Evaluation Results

Table 4.1 summarizes the results reported for the four evaluation datasets, separated by the underlying generator LLM. The model coupling CRAG with standard RAG is referred to simply as CRAG, and the model coupling CRAG with Self-RAG is referred to as Self-CRAG.

Table 4.1: Overall evaluation results on the test sets of four datasets.

Method	PopQA (Accuracy)	Bio (FactScore)	Pub (Accuracy)	ARC (Accuracy)
LMs trained with propriety data				
LLaMA2-c$_{13B}$	20.0	55.9	49.4	38.4
Ret-LLaMA2-c$_{13B}$	51.8	79.9	52.1	37.9
ChatGPT	29.3	71.8	70.1	**75.3**
Ret-ChatGPT	50.8	-	54.7	**75.3**
Perplexity.ai	-	71.2	-	-
Baselines without retrieval				
LLaMA2$_{7B}$	14.7	44.5	34.2	21.8
Alpaca$_{7B}$	23.6	45.8	49.8	45.0
LLaMA2$_{13B}$	14.7	53.4	29.4	29.4
Alpaca$_{13B}$	24.4	50.2	55.5	54.9
CoVE$_{65B}$	-	71.2	-	-
Baselines with retrieval				
LLaMA2$_{7B}$	38.2	78.0	30.0	48.0
Alpaca$_{7B}$	46.7	76.6	40.2	48.0
SAIL	-	-	69.2	48.4
LLaMA2$_{13B}$	45.7	77.5	30.2	26.0
Alpaca$_{13B}$	46.1	77.7	51.1	57.6
LLaMA2-hf-7b				
RAG	37.7	44.9	**9.1**	23.8
CRAG	39.8	47.7	**9.1**	25.8
Self-RAG*	29.0	32.2	0.7	23.9
Self-CRAG	**49.0**	**69.1**	0.6	**27.9**
SelfRAG-LLaMA2-7b				
RAG	40.3	59.2	39.0	46.7
CRAG	59.3	74.1	**75.6**	54.8
Self-RAG	54.9	81.2	72.4	67.3
Self-CRAG	**61.8**	**86.2**	74.8	67.2

The results lead to three key findings. First, CRAG consistently and significantly improves the performance of both RAG and Self-RAG: based on SelfRAG-LLaMA2-7b, CRAG outperformed plain RAG by 7.0% accuracy on PopQA, 14.9 FActScore on Biography, 36.6% accuracy on PubHealth, and 15.4% accuracy on Arc-Challenge, while Self-CRAG outperformed Self-RAG by 6.9% accuracy on PopQA, 5.0 FActScore on Biography, and 2.4% accuracy on PubHealth. Second, CRAG generalizes well across a variety of task types – short-form entity generation, long-form generation, and closed-set classification – demonstrating that the improvement is not confined to a single task format. Third, CRAG shows greater flexibility

when the underlying generator LLM is swapped: it remained competitive when moving from SelfRAG-LLaMA2-7b to the plain LLaMA2-hf-7b, whereas Self-RAG's performance dropped sharply (even falling below standard RAG on some benchmarks), because Self-RAG depends on an LLM that has been specially instruction-tuned to output its critic tokens, a requirement CRAG does not share.

4.2 Ablation Study

To verify the contribution of each design choice, the authors performed ablation studies on the PopQA dataset.

Table 4.2: Ablation study for removing each single action on the PopQA dataset in terms of accuracy.

	LLaMA2-hf-7b	SelfRAG-LLaMA2-7b
CRAG	54.9	59.8
w/o. Correct	53.2	58.3
w/o. Incorrect	54.4	59.5
w/o. Ambiguous	54.0	59.0
Self-CRAG	49.0	61.8
w/o. Correct	43.6	59.6
w/o. Incorrect	47.7	60.8
w/o. Ambiguous	48.1	61.5

Removing any one of the three actions – Correct, Incorrect, or Ambiguous – caused a measurable drop in performance, confirming that each action contributes to the robustness of generation. A similar ablation was performed on the individual knowledge-utilization operations.

Table 4.3: Ablation study for removing each knowledge utilization operation on the PopQA dataset in terms of accuracy.

	LLaMA2-hf-7b	SelfRAG-LLaMA2-7b
CRAG	54.9	59.8
w/o. refinement	49.8	54.2
w/o. rewriting	51.7	56.2
w/o. selection	50.9	58.6
Self-CRAG	49.0	61.8
w/o. refinement	35.9	52.2
w/o. rewriting	37.2	58.4
w/o. selection	24.9	57.9

Removing document refinement (feeding raw retrieved documents directly to the generator, as most existing RAG systems do), removing query rewriting (searching the web with the raw question instead of extracted keywords), or removing knowledge selection (using all searched web content without filtering) all degraded performance, showing that each of these refinement steps meaningfully improves how retrieved and searched knowledge is utilized.

4.3 Accuracy of the Retrieval Evaluator

Table 4.4: Evaluation of our retrieval evaluator and ChatGPT on the PopQA dataset.

	Accuracy
Our Retrieval Evaluator (T5-based)	84.3
ChatGPT	58.0
ChatGPT-CoT	62.4
ChatGPT-few-shot	64.7

The lightweight, purpose-trained T5-based evaluator clearly outperformed the much larger ChatGPT model across all prompting strategies (direct, chain-of-thought, and few-shot), demonstrating that a small, specifically fine-tuned evaluator is more reliable for this judgment task than a general-purpose LLM used off the shelf.

4.4 Robustness to Retrieval Performance and Web Search Contribution

The authors also studied how generation performance changes as retrieval quality itself degrades, by randomly removing a portion of accurate retrieval results to simulate a low-quality retriever.

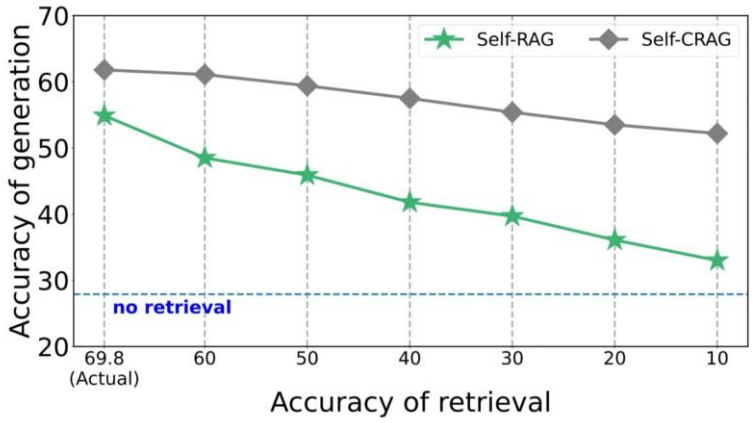


Figure 4.1: The generation performance of Self-RAG and Self-CRAG given different retrieval performance on the PopQA dataset with SelfRAG-LLaMA2-7b

As retrieval performance dropped, the performance of both Self-RAG and Self-CRAG declined, but Self-CRAG's performance degraded more gracefully than Self-RAG's, indicating superior robustness to poor retrieval. Separately, to confirm that CRAG's gains come from the self-correction mechanism rather than simply from the extra web knowledge, both RAG and Self-RAG were also consistently supplemented with the same web search knowledge used in CRAG.

Table 4.5: Comparison results between CRAG, Self-CRAG and RAG, Self-RAG with the same input in terms of accuracy.

	LLaMA2-hf-7b	SelfRAG-LLaMA2-7b
PopQA		
CRAG	54.9	59.8
RAG	50.5	52.8
RAG w. web	52.2	53.8
Self-CRAG	49.0	61.8
Self-RAG	29.0	54.9
Self-RAG w. web	24.9	57.9

While this supplementation gave a modest improvement in most cases, CRAG and Self-CRAG still significantly outperformed these web-augmented baselines, confirming that the self-correction mechanism – not merely the addition of web data – is the primary source of improvement.

4.5 Computational Overhead

Table 4.6: Computational overhead of RAG, CRAG, Self-RAG and Self-CRAG.

	TFLOPs per token	executing time(s)
RAG	26.5	0.363
CRAG	27.2	0.512
Self-RAG	26.5∼132.4	0.741
Self-CRAG	27.2∼80.2	0.908

Using FLOPs prediction formulas for the generation phase, CRAG adds only a small increase in per-token compute over standard RAG (27.2 vs. 26.5 TFLOPs per token), while measured end-to-end execution time on PopQA rose modestly from 0.363s to 0.512s per instance. This confirms that the self-correction mechanism is genuinely lightweight and can be added to existing RAG pipelines without imposing prohibitive computational cost.

5. CONCLUSION

This paper has examined Corrective Retrieval-Augmented Generation (CRAG), a framework that directly confronts a long-standing weak point of RAG systems: their vulnerability to poor-quality retrieval. By introducing a lightweight, dedicated retrieval evaluator to judge the relevance of retrieved documents, and by discriminately triggering one of three corrective actions – Correct, Incorrect, or Ambiguous – CRAG allows a RAG pipeline to recognize when its retrieved knowledge cannot be trusted and to seek out better information (via web search) rather than blindly using what was returned. The accompanying decompose-then-recompose knowledge refinement strategy further ensures that only the most relevant fragments of a document, whether retrieved locally or found on the web, are passed on to the generator.

Experimental results across four diverse datasets – PopQA, Biography, PubHealth and Arc-Challenge – show that CRAG consistently and substantially improves both standard RAG and the more sophisticated Self-RAG, regardless of the underlying generator LLM. Ablation studies confirm that every individual component of the design (the three actions, as well as document refinement, query rewriting and knowledge selection) contributes meaningfully to the overall gains, and the evaluator itself was shown to be considerably more accurate than prompting a general-purpose LLM such as ChatGPT for the same judgment task. Importantly, all of this is achieved at only a modest increase in computational cost, underlining CRAG's practicality as a plug-and-play addition to existing RAG-based systems.

The authors acknowledge one key limitation: CRAG still relies on an external, separately fine-tuned retrieval evaluator, which adds a training and maintenance burden. They identify eliminating this external evaluator – for instance, by equipping the generator LLM itself with better built-in retrieval-evaluation capability – as an important direction for future work. More broadly, as LLM-based systems are increasingly deployed in knowledge-intensive, high-stakes settings, techniques like CRAG that make retrieval-augmented generation self-aware of its own knowledge quality are likely to remain an important building block for building trustworthy and factually reliable AI systems.

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